

Ge Yan

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RESEARCH INTERESTS

Trustworthy machine learning and responsible AI, with a focus on mechanistic interpretability, uncertainty quantification, and robust prediction. Recent work uses representation engineering and model editing to understand and improve LLM reasoning, reflection, and tool use.

EDUCATION

University of California San Diego, Computer Science and Engineering Sep 2023 – Present
Ph.D. student, Trustworthy ML Lab. Advisor: Prof. Tsui-Wei (Lily) Weng.

University of California San Diego, Electrical and Computer Engineering Sep 2021 – Mar 2023
M.S. in Machine Learning and Data Science. GPA: 3.95/4.00.

Peking University, School of Mathematical Sciences Sep 2017 – Jun 2021
B.S. in Information and Computing Science. GPA: 3.484/4.00.

INDUSTRY EXPERIENCE

Machine Learning Engineer Intern, Stripe, South San Francisco, CA Jun 2026 – Sep 2026
Agent Design and Development Team

- Evaluated common uncertainty metrics for internal LLM-as-a-judge systems.
- Collected a dataset of agent traces for evaluation; compared models, prompting strategies, and other hyperparameters.

Applied Scientist Intern, Amazon, San Diego, CA Jun 2024 – Aug 2024

- Developed a machine-generated text detector using style embeddings; improved robustness to adversarial inputs and accelerated the existing model.
- Delivered a lightweight, robust detection model adopted in an internal tool.

Data Scientist Intern, DiDi Technology, Beijing, China Jun 2023 – Aug 2023

- Developed and maintained causal inference tools to guide marketing decisions, including Bayesian Structural Time Series for counterfactual prediction.
- Supported internal analyst teams with a tool used more than 100 times per month.

SELECTED RESEARCH EXPERIENCE

ReflCtrl: Efficient control of LLM reflection

- Developed stepwise representation steering to control reflection frequency, saving up to 33.6% of reasoning tokens while largely preserving accuracy in evaluated settings.

VLG-CBM: Faithful concept bottleneck models

- Co-developed vision-language guidance and the Number of Effective Concepts metric to improve concept faithfulness and control information leakage across five benchmarks.

Provably robust conformal prediction

- Developed theoretical analysis and two methods for prediction sets robust to adversarial inputs; reduced baseline inefficiency by up to 48.80% on ImageNet.

TECHNICAL SKILLS

Python, C, PyTorch; deep learning experimentation, statistical evaluation, and causal inference. Strong foundation in linear algebra, probability, mathematical analysis, and real analysis.

PUBLICATIONS

Conference and journal papers

1. **ReflCtrl: Controlling LLM Reflection Efficiently via Representation Engineering**
Ge Yan, Chung-En Sun, Linbo Liu, Tsui-Wei Weng. COLM 2026 (accepted). Also NeurIPS 2025 MI Workshop Spotlight.
2. **Steer2Edit: From Activation Steering to Component-Level Editing**
Chung-En Sun, Ge Yan, Zimo Wang, Tsui-Wei Weng. NeurIPS 2026 Main Track (accepted, poster).
3. **LLM Agents Already Know When to Call Tools — Even Without Reasoning**
Chung-En Sun, Linbo Liu, Ge Yan, Zimo Wang, Tsui-Wei Weng. NeurIPS 2026 Main Track (accepted, poster).
4. **Distance Marching for Generative Modeling**
Zimo Wang, Ishit Mehta, Haolin Lu, Chung-En Sun, Ge Yan, Tsui-Wei Weng, Tzu-Mao Li. NeurIPS 2026 Main Track (accepted, poster).
5. **ReFINE: A Framework for Trustworthy Large Reasoning Models with Reliability, Faithfulness, and Interpretability**
Chung-En Sun, Ge Yan, Akshay Kulkarni, Tsui-Wei Weng. COLM 2026 (accepted).
6. **Multimodal Concept Bottleneck Models**
Tongqing Shi, Ge Yan, Tuomas Oikarinen, Tsui-Wei Weng. TMLR 2026 (accepted with minor revision).
7. **Beyond Top Activations: Efficient and Reliable Crowdsourced Evaluation of Automated Interpretability**
Tuomas Oikarinen, Ge Yan, Akshay Kulkarni, Tsui-Wei Weng. CVPR 2026.
8. **ThinkEdit: Interpretable Weight Editing to Mitigate Overly Short Thinking in Reasoning Models**
Chung-En Sun, Ge Yan, Tsui-Wei Weng. EMNLP 2025.
9. **Evaluating Neuron Explanations: A Unified Framework with Sanity Checks**
Tuomas Oikarinen, Ge Yan, Tsui-Wei Weng. ICML 2025.
10. **Interpretable Generative Models through Post-hoc Concept Bottlenecks**
Akshay Kulkarni, Ge Yan, Chung-En Sun, Tuomas Oikarinen, Tsui-Wei Weng. CVPR 2025.
11. **VLG-CBM: Training Concept Bottleneck Models with Vision-Language Guidance**
Divyansh Srivastava*, Ge Yan*, Tsui-Wei Weng. NeurIPS 2024. * Equal contribution.
12. **Provably Robust Conformal Prediction with Improved Efficiency**
Ge Yan, Yaniv Romano, Tsui-Wei Weng. ICLR 2024.

Workshop papers and preprints

13. **Faithful and Stable Neuron Explanations for Trustworthy Mechanistic Interpretability**
Ge Yan, Tuomas Oikarinen, Tsui-Wei Weng. NeurIPS 2025 Mechanistic Interpretability Workshop.
14. **RAT: Boosting Misclassification Detection Ability without Extra Data**
Ge Yan, Tsui-Wei Weng. arXiv preprint, 2025.

ACADEMIC SERVICE AND TALKS

Area Chair: ICLR 2026 Trustworthy AI Workshop; CVPR 2026 TRUE-V Workshop.

Reviewer: ICLR 2025, 2026; NeurIPS 2025, 2026; ICML, CVPR, ECCV, and COLM 2026.

Tutorial Co-organizer: Principled Interpretability in Vision Models, CVPR 2026.

Invited Talk: Faithful Interpretation for Deep Networks via Human Understandable Concepts, EnCORE Workshop on Interpretability in Modern AI, 2026.